

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology and Engineering

Second Semester of M. Tech. Examination (CE)

May 2013

CE710.01 High Performance Computing

Date: 13.05.2013, Monday Time: 10:00 a.m. To 01:00 p.m. Maximum Marks: 70

Instructions:

- (1) All questions are compulsory.
- (2) Indicate clearly, the option you attempt along with its respective question Number.
- (3) Figures to right indicate marks.
- (4) Rough work is done in the last page of main supplementary; don't write anything on the question paper.

SECTION-I

Q-1

- (a) Use examples to explain the following terms on cluster job management system. [04]
- (1) Cluster processes, local processes, and kernel processes
 - (2) Dedicated mode, space-sharing mode and timesharing mode
- (b) What are the various way to connect cluster nodes? Explain any of them in brief. [03]
- (c) Explain layered grid service protocols and their relationship with the internet service protocols. [03]
- (d) What are the pre-requisites necessary for running an Open MPI job? [01]

Q-2 Answer the following questions.

- (a) How to configure HA (High Availability) using two Linux machines running a Heartbeat? Explain with proper example. [05]
- (b) Differentiate between dedicated server process and shared server process with an example. [04]
- (c) List and explain the classification of cluster node into different layers (from viewpoint of a process P). [03]

OR

- (c) Give the comparison of Contention Aware List Scheduling Heuristics for Cluster Computing. [03]

Q-3 Answer the following questions. (Any Three)

- [12]
- (a) Explain the relationships among clustering middleware at the job management, programming and implementation levels.
- (b) How the cluster provide availability support for any server node? Illustrate with an example.
- (c) Load-sharing algorithms do not need to exchange periodic state information among all nodes in the system. Justify.
- (d) Under which situations can the process be complete immediately blocked prior to migration?

SECTION-II

Q-4 Answer the following questions.

- (a) Which are the basic cluster components based on the communication software. Explain any one in brief. [02]
- (b) Differentiate between user-level and kernel-level thread. [02]
- (c) How to enable communication over the memory channel network? [03]
- (d) Explain any four typical techniques to optimize communication (interprocess communication)? [04]

Q-5 Answer the following questions.

- (a) For any pipelined CPU, is it mandatory for each of its machine instruction to be of identical length? Justify your answer with proper reasoning. [03]
- (b) Which are the basic steps to be performed in a transaction-based system that uses Network RAM? Explain in detail. [04]
- (c) Which are the various approaches to define the structure of the address space supported by the DSM system?. [05]

OR

- (c) In Network RAM, failure of workstation and network causes unavailability pages. What are the solutions to overcome that situation? Which one is very efficient among those solutions? [05]

Q-6 Answer the following questions.

- (a) Draw and explain architecture of network paging system. [04]
- (b) Explain about cluster computer architecture with proper diagram. [04]
- (c) What do you mean by virtualization? Compare the HyperV and Xen virtualization softwares. [04]